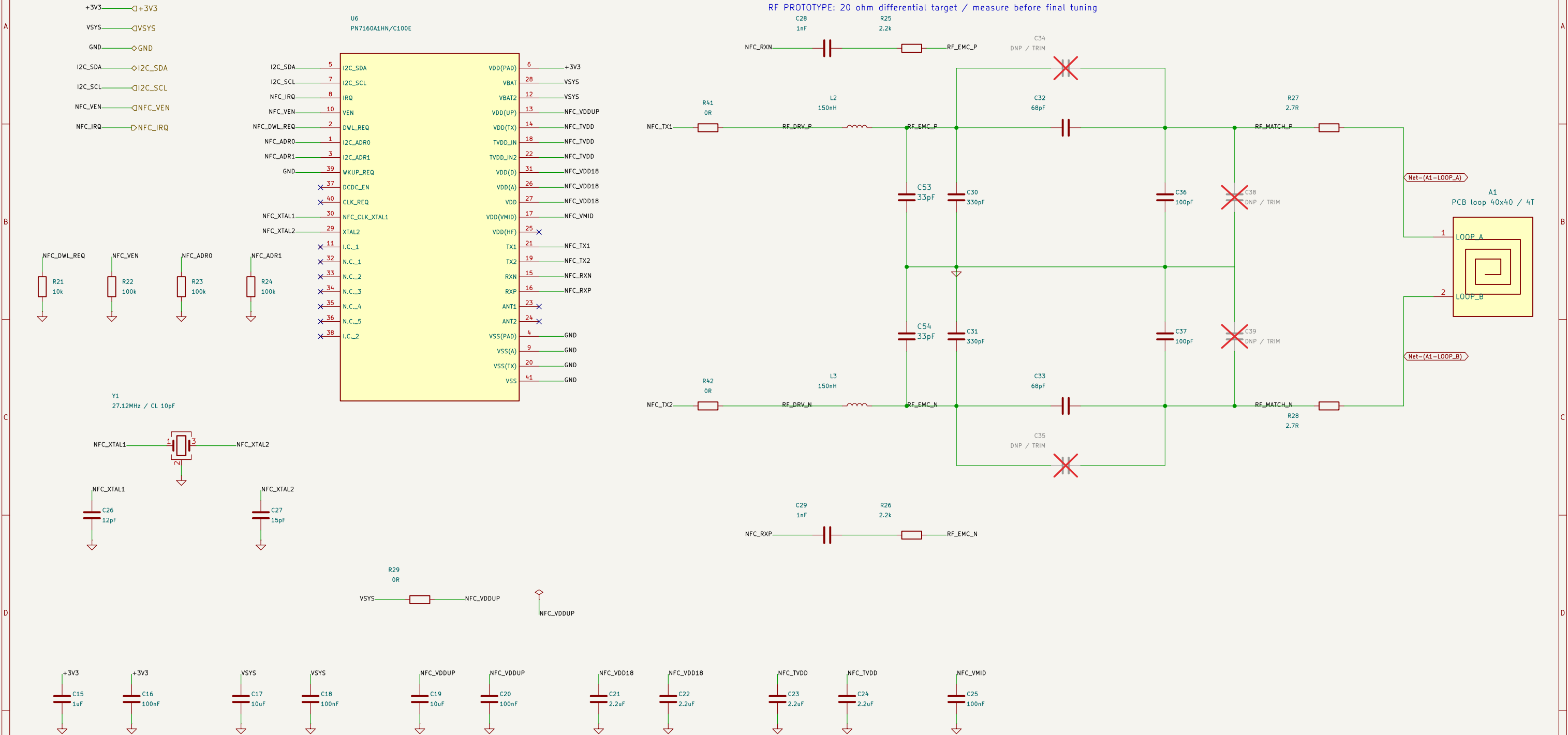


PN7160 / I2C host + balanced NFC antenna matching



Pin 14 -> 18 + 22: internal TX-LDO output, not an external supply.
Pin 31 -> 26 + 27: internal 1.8 V rail; decouple only.
VSYS powers VBAT/VBAT2 and VDD_UP; VDD_PAD uses +3V3.
ADR1:0 = 00 => 0x28. WKUP_REQ low: wake over I2C.
VEN defaults low. DWL_REQ test access via its labelled net.
ANT1/ANT2/VDHF unused: configure normal RF detection.
Y1: NDK CS06346 27.12 MHz, CL 10pF; C26=12pF / C27=15pF.
C21 at pin 26, C22 at 27; C23 at 18, C24 at 22: 2.2uF each.

A1: 40x40 mm, 4T, 0.50/0.30 mm; estimated 1.56 uH. No GND tap.
C34/C35, C38/C39: paired 0805 COG tuning pads, DNP.
C30/C31 + C53/C54: 330 + 33 pF = 363 pF EMC shunt per leg.
Lift R41/R42 and R25/R26 to isolate RF tree for passive VNA tests.
Lift R27/R28 for bare-coil measurement. NEVER power a connected VNA.
Start TVDD <= 3.3 V: 20 ohm target trades field strength for current margin.
See docs/nfc-antenna.md for calculations, layout rules and tuning sequence.